Laksh Gupta

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Master of Science in Astrophysics student at Argelander-Institut für Astronomie, University of Bonn. Currently working on understanding the effect of the environment on galaxy evolution. This work is being supervised by Prof. Dr. Cristiano Porciani and Dr. Mohammadreza Ayromlou. Supported through the **full scholarship** of Bonn-Cologne Graduate School of Physics and Astronomy.

Education

Master of Science in Astrophysics, *University of Bonn, Germany* October 2025 - October 2027

- Expected GPA - 1.4/4.0 (highest: 1.0, lowest: 4.0)
- **Relevant Coursework**: Theoretical Astrophysics (General Relativity & Fluid Dynamics), Cosmology, Physics of Dense Stellar Systems, X-ray Astronomy, Advanced Laboratory, Galaxy Clusters, Stellar Astrophysics, Interstellar Medium, Astrophysics of Galaxies

Bachelor of Science (Honors) in Physics, *Ahmedabad University, India* August 2020 - December 2024

- GPA - 3.3/4.0 (highest: 4.0, lowest: 1.7)
- Minor in Mathematics
- **Relevant Coursework**: Quantum Mechanics, Atomic & Nuclear Physics, Condensed Matter Physics, Plasma Physics, Nonlinear Dynamics, Computational Mathematics, Quantum Computing, Complex Analysis, Mathematical Statistics, Advanced Writing

Research Experience

Master's Thesis, Large Scale Structure & Galaxy Formation Group October 2026 - October 2027
Supervisors: Prof. Dr. Cristiano Porciani & Dr. Mohammadreza Ayromlou *Argelander-Institut für Astronomie*

- Aim to implement cold gas stripping mechanism to L-GALAXIES.

Research Intern, Large Scale Structure & Galaxy Formation Group March 2026 - June 2026
Supervisors: Prof. Dr. Cristiano Porciani & Dr. Mohammadreza Ayromlou *Argelander-Institut für Astronomie*

- Worked on a 6-week project to study the environmental dependence of galaxy properties beyond the virial radius of the halo at $z = 0$ using the TNG100-1 simulation.
- Built a `Python` pipeline to kinematically classify galaxies into categories such as infalling or outfalling with respect to their local background environment. Ram pressure and specific star-formation rate estimates were then used to quantitatively estimate and understand quenching.
- **This work is being written up for publication**.

Research Intern, Astronomy & Astrophysics Group June 2025 - August 2025
Supervisor: Prof. Blesson Mathew *Christ University, Bengaluru, India*

- In collaboration with Arun Roy (Indian Institute of Astrophysics, Bengaluru) and Shridharan Baskaran (Tata Institute of Fundamental Research, Mumbai), I worked on modelling a Herbig Be star with boundary layer accretion mechanism in RADMC3D to understand forbidden [Ne II] and [Ne III] lines.

Summer Research Fellow April 2025 - June 2025
Supervisor: Prof. Blesson Mathew *Science Academies' Summer Research Fellowship*

- Conducted a detailed theoretical and observational review on accretion mechanisms in Herbig Ae/Be stars, identifying a switch from magnetospheric to boundary layer accretion.

- Prepared a Python pipeline to produce the largest and most complete catalogue of identified Herbig Ae/Be stars from literature.
- Initiated modeling efforts using RADMC3D to study switch in accretion mechanism in Herbig Ae/Be stars, generating SEDs that match the inclination effects and silicate features in Herbig-like systems.

Research Intern, Astronomy & Astrophysics Group May 2024 - October 2025
Supervisor: Prof. Samyaday Choudhury *Ahmedabad University*

- In collaboration with Dr. Annalisa Calamida to refine the star count-crossing time analysis technique developed during my undergraduate thesis.
- A **pipeline** in `Python` was created for the analysis. This is unique as it provides star counts and evolutionary times using five parameters (age, mass, envelope thickness, membership probability, and distance modulus), making the predictions more robust.
- This work has been published in *The Astrophysical Journal* (link).

Bachelor's Thesis - *Study of White Dwarfs in the Globular Cluster NGC2808* (link) July 2023 - May 2024
Supervisor: Prof. Samyaday Choudhury *Ahmedabad University*

- Performed star count-crossing time analysis of different evolutionary phases using photometric data of NGC2808 from the HUGS survey.
- Overplotted various BaSTI models (White Dwarf Cooling Models, Isochrones and Evolutionary Tracks) on NUV-optical Color-Magnitude Diagrams to calculate crossing times across different evolutionary phases.
- Created a Python pipeline to systematically analyze stellar populations (White Dwarf Stars, Red Giant Branch Stars, Main Sequence Turnoff Stars) of NGC2808, potentially applicable to other globular clusters.
- *Our results indicate that the White Dwarf Sequence in NGC2808 has no bimodality up to 24 magnitude in F275W.*

Publications

Gupta, L., Choudhury, S., Calamida, A., Johnson, C. and Nardiello, D., 2025. *An Excess of Luminous White Dwarfs in the Peculiar Galactic Globular Cluster NGC 2808*. *The Astrophysical Journal*, 994, 97

Gupta, L., Ayromlou, M., Porciani, C., *in preparation*

Skills

Astronomy	BaSTI and MIST Models (White Dwarf Cooling Models, Isochrones, Evolutionary Tracks), Optical and Near-IR Hubble Data Analysis, Color-Magnitude Diagram Analysis, Reddening Estimation, IllustrisTNG Simulation Data Analysis, HDF5
Tools	RADMC3D, VizieR, xMatch, TOPCAT, Tracker
Collaborative Tools	TEX, `VSCode`, `git` (branching, resolving conflicts)
Programming Languages	`Python` (`NumPy`, `SciPy`, `Pandas`, `Astropy`, `Shapely`, `Seaborn`), `MATLAB`, `Java`, `R`, `C`
Communication	Fluent in English (IELTS 7 band), Fluent in Hindi

Awards & Achievements

- Awarded the Bonn-Cologne Graduate School of Physics and Astronomy Scholarship 2025.
- Awarded the Indian Science Academies' Summer Research Fellowship 2025.
- Qualified IIT-Joint Admission test for Masters (IITJAM) 2025 (top 6 percentile in the country).

Conferences

- **Talk** at Joint L-Galaxies/GAEA workshop, INAF, Trieste, 30th November - 3rd December 2026 (Upcoming)
- *Excess of Luminous WDs in the Globular Cluster NGC 2808* **poster** (link) at 24th European Workshop on White Dwarfs, Institute of Science & Technology, Austria, 3rd - 7th August 2026
- *Study of White Dwarfs in NGC 2808 using Near-Ultraviolet and Optical data from the Hubble Space Telescope* **poster** (link) at 43rd Meeting of the Astronomical Society of India, NIT Rourkela, 15th - 19th February 2025

Tutor Experience

- Tutor for *Cosmology* at Argelander-Institut für Astronomie (October 2026 - March 2027)
- Mentored a summer intern in the Astronomy & Astrophysics Group at Ahmedabad University (May 2024 - June 2024)
- Tutor for *Atomic & Nuclear Physics* at Ahmedabad University (January 2024 - May 2024)
- Tutor for *Advanced Writing* at Ahmedabad University (August 2021 - December 2021)

Attended Schools

- *From Particle Physics to Astrophysics - Cold Dark Matter and Beyond*, 56th Saas-Fee Advanced **Course**, Saas-Fee, Switzerland, 24th January - 29th January 2027 (Upcoming)
- National **School** on Space Radiation, Manipal Center Natural Sciences, MAHE, Manipal, 16th - 20th September 2024
- **School** on Solar Physics Physical Research Laboratory, Ahmedabad, 2nd - 6th September 2024

Leadership and Volunteering Activities

- Member of the Diversity, Equity and Inclusion committee at the Argelander-Institut für Astronomie, University of Bonn (April 2026 - Present)
- Part of the *Study-Buddy Program* at the International Office of the University of Bonn (September 2025 - September 2026)
- Organized a *Math Fest* at Ahmedabad University (November 2023)
- Founded *Ramanujan Math Club* promoting student problem-solving and mathematics at Ahmedabad University (August 2023 - May 2024)
- Represented Physics at Ahmedabad University's *Career Development Centre* (May 2023 - May 2024)
- *Student Mitr* - Mentored 15 incoming students to acquaint them with Ahmedabad University's culture (August 2021 - April 2022)
- *Junior Manager* Outgoing Social Sector - Association Internationale des Étudiants en Sciences Économiques et Commerciales (AIESEC) in Ahmedabad (August 2021 - October 2021)
- Volunteered at Prabhat Foundation during the COVID-19 pandemic (May 2021 - July 2021)